Experiment 4

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Problem: Multiply two integers without using the * operator using repeated addition.

Incorrect Algorithm (Sample)

1. Read num1 and num2
2. prod = 0
3. Repeat num2 times
   prod = prod + num1
4. Print prod
5. Stop

Mistakes in the Above Algorithm

• Does not handle negative numbers.
• Fails when either input is 0.
• Does not use the minimum number of additions.
• Does not determine the correct sign of the answer.

Correct Algorithm

1. Start
2. Read num1 and num2.
3. If num1==0 or num2==0, print 0 and Stop.
4. sign = 1.
5. If num1<0, num1=-num1, sign=-sign.
6. If num2<0, num2=-num2, sign=-sign.
7. If num1<num2, swap(num1,num2).
8. prod=0.
9. Repeat num2 times:
   prod = prod + num1
10. If sign<0, prod=-prod.
11. Print prod.
12. Stop.

Incorrect Flowchart (Sample)

Start → Input num1,num2 → prod=0 → Repeat num2 times → prod=prod+num1 → Print prod → Stop

Mistakes in Flowchart

• No decision for zero values.
• No handling of negative values.
• No swapping to reduce additions.
• Missing sign correction before output.

Correct Flowchart (Text Form)

Start → Input num1,num2 → Any input zero? → Yes: Print 0 → Stop.
No → Determine sign → Convert numbers to positive → Is num1<num2? → Yes: Swap values → Initialize prod=0 → Loop num2 times adding num1 → Apply sign → Print product → Stop.

Verification Table

Input	Expected	Minimum Additions
12,3	36	3
3,12	36	3
-6,5	-30	5
6,-5	-30	5
-9,-4	36	4
-4,-9	36	4
0,8	0	0
8,0	0	0
0,-7	0	0
-7,0	0	0
0,0	0	0
1,-7	-7	1
-1,-9	9	1